VECTOR { unknown } Echelon Seas

Arizona State University

A game about linear algebra, vectors, matrices and pirates

Don't have time right now? Want to play at your hotel/home?

Play online at:

tinyurl.com/EchelonSeas

Join our expo session using code: **SGSC2021**
or make your own account to save your progress!

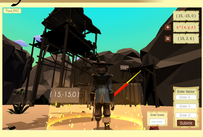
The Island Hub

Trapped on an island, your ship is torn to pieces. Here you can access the other puzzle stages in any order you choose, explore your ship or review your records.

Each completed puzzle rewards the player with a piece of the ship. Though you can always play the same stage again to try for a better score, a faster time, or in less moves.

Stage1: Vector Displacement

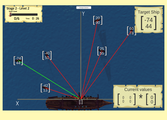
Stage1 illustrates vectors as they are commonly taught in school. Using simple vector math the player can build bridges to the next question.
This stage is excellent for getting a first feel control of the camera and player movement.



Stage2: Matrix Multiplication

Stage2 introduces matrix transformation by providing selectable options of cannons and cannonballs.

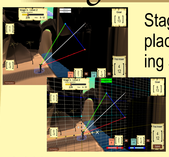
The player aims their cannon by choosing a matching pair of 2x2 matrix and an vector to match the realworld position of the enemy ship.



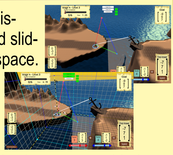
Stage3: Vector Spans

Stage3 provides a different way to think about vector spans. It asks the player to navigate a ball through a maze using by altering the vectors in the span. The player tilts the resulting plane of possibilities provided by the vectors in the span.

Stage4: Scalar Vector Displacement



Stage4 uses a more advanced version of vector displacement. By selecting a pair of vector cards and sliding a scalar it illustrates where the vectors are in space. The player must find a matching pair and scalar values to aim a grapple gun at the anchor.



Addtional vector visualization tools such as a graph, axis lines and projections are provided to help the player.

Organizations and Developers

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Built from Educational Research
The game "VectorUnknown: Echelon Seas" is based on the previous 2D iteration "VectorUnkown" from the Inquiry Oriented Linear Algebra curriculum.

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